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S2

Started on	Thursday, 1 October 2026, 9:32 AM
State	Finished
Completed on	Thursday, 1 October 2026, 9:33 AM
Time taken	46 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 // Function to find and print the intersection of two sorted arrays
5 void printIntersection(int* arr1, int n1, int* arr2, int n2) {
6     int i = 0, j = 0;
7     int first_print = 1;
8
9     // Traverse both sorted arrays simultaneously using two pointers
10    while (i < n1 && j < n2) {
11        if (arr1[i] < arr2[j]) {
12            i++; // Advance the first array pointer
13        }
14        else if (arr2[j] < arr1[i]) {
15            j++; // Advance the second array pointer
16        }
17        else {
18            // Found a common element matching the intersection
19            if (!first_print) {
20                printf(" ");
21            }
22            printf("%d", arr1[i]);
23            first_print = 0;
24
25            i++;
26            j++;
27        }
28    }
29    printf("\n");

```

```
30 }
31
32 int main() {
33     int t;
34     if (scanf("%d", &t) != 1) return 0;
35
36     // Process all T test cases
37     while (t--) {
38         int n1, n2;
39
40         // Read N1 and the elements of the first array
41         if (scanf("%d", &n1) != 1) break;
42         int* arr1 = (int*)malloc(n1 * sizeof(int));
43         for (int i = 0; i < n1; i++) {
44             if (scanf("%d", &arr1[i]) != 1) break;
45         }
46
47         // Read N2 and the elements of the second array
48         if (scanf("%d", &n2) != 1) {
49             free(arr1);
50             break;
51         }
52         int* arr2 = (int*)malloc(n2 * sizeof(int));
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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